

# **LipidMem-BigSMILES (LiMBS): A Standardized Representation Language for Lipid Membrane Systems**

**LiMBS1.0**

User Manual

## 1. Overview

LiMBS1.0 is a line notation for describing lipid membrane simulation systems in a human-readable, machine-parseable format. It encodes membrane geometry, lipid composition, environmental conditions, and molecular structure in a single self-contained string.

LiMBS supports atomistic (AA), coarse-grained (CG), or super coarse-grained (SCG) representations. The LiMBS parser validates the notation and can generate canonical LiMBS and structured JSON output. Separate LiMBS\_INSANE and LiMBS\_TS2CG interfaces translate compatible CG specifications into inputs for the external INSANE and TS2CG membrane-building software, respectively.

### 1.1 Supported Resolutions

| Field | Description                                    | Example  |
|-------|------------------------------------------------|----------|
| AA    | Atomistic — heavy-atom SMILES notation         | rsln:AA  |
| CG    | Coarse-grained — bead mapping notation         | rsln:CG  |
| SCG   | Super coarse-grained — semantic group notation | rsln:SCG |

### 1.2 Supported Geometries

| Field   | Description                        | Example      |
|---------|------------------------------------|--------------|
| planar  | Flat bilayer — use with INSANE     | type:planar  |
| vesicle | Spherical vesicle — use with TS2CG | type:vesicle |

### 1.3 Dual-Block Architecture

Every LiMBS string consists of two blocks separated by the |++| delimiter:

```
[ System Block ]
|++|
[ Species Block ]
```

System Block - describes the membrane geometry, composition summary, box, solvent, and salt.

Species Block - defines the chemical structure of each species at the chosen resolution.

## 2. System Block

The system block is a sequence of fields separated by double-pipe delimiters (||). The salt: field is optional. When present, it is normally the final system field before the |++| separator.

### 2.1 Field Reference

| Field    | Description                 | Example  |
|----------|-----------------------------|----------|
| LiMBS1.0 | Version token must be first | LiMBS1.0 |
| rsln:    | Resolution (AA, CG, or SCG) | rsln:CG  |

| Field    | Description                                                                                                                                                                                   | Example                    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| lsc:     | Number of lipid species                                                                                                                                                                       | lsc:2                      |
| uls:     | Number of upper leaflet species                                                                                                                                                               | uls:2                      |
| lls:     | Number of lower leaflet species                                                                                                                                                               | lls:1                      |
| {...}    | Lipid metadata block                                                                                                                                                                          | {DPPC:tails=16:0-16:0,...} |
| leaflet: | Leaflet composition — count or ratio. For planar CG systems, both leaflet: count and leaflet: ratio is supported. For CG vesicles, the current LiMBS-TS2CG interface requires leaflet: ratio. | leaflet:count{...}         |
| type:    | Membrane geometry                                                                                                                                                                             | type:planar                |
| box:     | Box dimensions in nm                                                                                                                                                                          | box:[10.0,10.0,8.0]nm      |
| sol:     | Solvent species name                                                                                                                                                                          | sol:W                      |
| salt:    | Salt concentration and ion type                                                                                                                                                               | salt:0.15M NaCl            |

## 2.2 Formatting Rules

**CRITICAL:** No space between field names and their values. Write rsln:CG not rsln: CG. This applies to every field without exception.

- Every header field except the final salt line must end with ||
- The lipid metadata block {...} must appear before the leaflet definition
- Metadata entries for multiple lipids are separated by semicolons inside one {...} block
- Do not split metadata into multiple {...} blocks

## 2.3 Planar System-Block Example (CG) — Count Mode

```
LiMBS1.0 ||
rsln:CG ||
lsc:1 ||
uls:1 ||
lls:1 ||
{DPPC:tails=16:0-16:0,head=PC,charge=0.0} ||
leaflet:count{-u{DPPC:56}-l{DPPC:56}} ||
type:planar ||
box:[10.0,10.0,8.0]nm ||
sol:W ||
salt:0.15M NaCl
```

## 2.4 Planar System-Block Example (CG) — Ratio Mode

Leaflet:ratio is also valid for type: planar. See Section 4 for how ratio mode is resolved for planar systems.

```

LiMBS1.0 ||
rsln:CG ||
lsc:2 ||
uls:2 ||
lls:2 ||
{POPC:tails=16:0-18:1,head=PC,charge=0.0;CHOL:tails=sterol,head=CHOL,charge=0.0} ||
leaflet:ratio{-u{POPC:0.70,CHOL:0.30}-l{POPC:0.70,CHOL:0.30}} ||
type:planar ||
box:[15.0,15.0,10.0]nm ||
sol:W ||
salt:0.15M NaCl

```

## 2.5 Vesicle System-Block Example (CG)

```

LiMBS1.0 ||
rsln:CG ||
lsc:2 ||
uls:2 ||
lls:2 ||
{POPC:tails=16:0-18:1,head=PC,charge=0.0;CHOL:tails=sterol,head=CHOL,charge=0.0} ||
leaflet:ratio{-u{POPC:0.70,CHOL:0.30}-l{POPC:0.70,CHOL:0.30}} ||
type:vesicle ||
box:[20.0,20.0,20.0]nm ||
sol:W ||
salt:0.15M NaCl

```

## 3. Lipid Metadata

The metadata block encodes biological lipid identity independent of resolution or force field.

### 3.1 Format

```
{LIPID:tails=<tail_composition>,head=<headgroup>,charge=<net_charge>}
```

For multiple lipids, separate entries with semicolons inside a single block:

```
{DPPC:tails=16:0-16:0,head=PC,charge=0.0;POPE:tails=16:0-18:1,head=PE,charge=0.0}
```

### 3.2 Tail Composition Notation

| Field     | Description                          | Example         |
|-----------|--------------------------------------|-----------------|
| 16:0      | Palmitoyl — 16 carbons, saturated    | tails=16:0      |
| 18:1      | Oleoyl — 18 carbons, one double bond | tails=18:1      |
| 16:0-18:1 | Two-chain lipid (e.g. POPC)          | tails=16:0-18:1 |
| sterol    | Cholesterol ring structure           | tails=sterol    |
| mixed     | Complex / heterogeneous (e.g. LPS)   | tails=mixed     |

### 3.3 Common Lipid Metadata Examples

```
DPPC:tails=16:0-16:0,head=PC,charge=0.0
POPC:tails=16:0-18:1,head=PC,charge=0.0
POPE:tails=16:0-18:1,head=PE,charge=0.0
POPG:tails=16:0-18:1,head=PG,charge=-1.0
POPS:tails=16:0-18:1,head=PS,charge=-1.0
POPA:tails=16:0-18:1,head=PA,charge=-1.0
CHOL:tails=sterol,head=CHOL,charge=0.0
LPS:tails=mixed,head=LPS,charge=-8.0
PVCL2:tails=16:0-18:1-16:0-18:1,head=CL,charge=-2.0
```

**NOTE:** Metadata records chain length and unsaturation degree only. Exact double-bond positions are specified inside the species block.

## 4. Leaflet Composition

The leaflet field defines which lipids populate the upper and lower leaflets, along with either their exact quantities (count mode) or their relative proportions (ratio mode). For planar CG systems, both leaflet: count and leaflet: ratio is supported. For CG vesicles, the current LiMBS-TS2CG interface requires leaflet: ratio.

### 4.1 Count Mode

```
leaflet:count{-u{DPPC:56}-l{DPPC:56}}
```

Each number is an absolute lipid count for that leaflet. Use count mode when the user wishes to specify explicit integer target populations for each leaflet, for example when reproducing the composition of a previously reported system or defining a fixed target composition. The final populations generated by INSANE should still be verified from the resulting topology.

Mixed asymmetric example:

```
leaflet:count{
-u{LPS:99}
-l{PVCL2:26,PMPE:104,PMPG:13,PVPE:104,PVPG:26}
}
```

### 4.2 Ratio Mode

```
leaflet:ratio{-u{POPC:0.50,CHOL:0.30,POPE:0.20}-l{POPC:0.50,CHOL:0.30,POPE:0.20}}
```

Each number is a relative proportion (mol fraction) of that lipid within the leaflet; the values in each leaflet should sum to 1.0.

### 4.3 How Each Mode Is Resolved Downstream

LiMBS itself does not convert leaflet:ratio into lipid counts. It parses and validates the notation, then hands the leaflet data — counts or ratios, unchanged — to whichever exporter matches the system's geometry.

#### *Planar systems (LiMBS\_INSANE.py → INSANE)*

- In leaflet:count mode, LiMBS preserves the user-specified integer values and passes them to the LiMBS-INSANE interface. However, the final number of molecules in the INSANE-built

membrane is determined by the builder's packing procedure and should be verified from the generated topology.

- In leaflet:ratio mode, the -u/-l arguments receive the specified relative proportions, while the box: dimensions are passed as -x, -y, and -z. INSANE determines the final integer lipid populations during membrane packing.
- LiMBS does not calculate the final integer populations from area-per-lipid values. The builder-resolved populations are recorded in the resulting topology.

#### ***Vesicle systems (LiMBS\_TS2CG.py → TS2CG)***

- Ratio values are written into input.str as relative proportions in the [Lipids List] block (see 11.3) and are consumed directly by TS2CG's domain-based composition step during PCG.
- For vesicle systems, the current LiMBS\_TS2CG interface requires leaflet: ratio; leaflet: count is not supported by the current vesicle interoperability workflow.

**PRACTICAL IMPLICATION:** With leaflet: ratio on a planar system, the final number of lipids in the built membrane is an emergent result of the box size and INSANE's own packing rules. With leaflet: count, LiMBS passes the specified integer values through unchanged, but the builder-resolved populations should still be checked against the generated topology in either mode. Section 5 gives guidance on sizing the box for either mode.

#### **4.4 Choosing Count vs. Ratio for Planar Systems**

| <b>Situation</b>                                                                                                                                                                     | <b>Recommended mode</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Reproducing a specific published system, or a system where explicit integer target populations are known (e.g. matching a prior simulation, or comparing across replicas at fixed N) | count                   |
| Screening or comparing lipid compositions (mol %) where the box size, not the exact lipid number, is the controlling parameter                                                       | ratio                   |
| Building an asymmetric membrane where explicit integer target populations are known for each leaflet                                                                                 | count                   |
| Building a membrane where you know the desired mol fractions but want INSANE to determine how many lipids fit the box                                                                | ratio                   |

### **5. Box Size and System Dimension Guidance**

The box: field is one of the most common sources of a system that parses correctly but builds a poorly packed, over-compressed, or unnecessarily large membrane. This section gives practical rules of thumb for choosing box\_x, box\_y, and box\_z.

#### **5.1 Lateral Dimensions (box\_x, box\_y) Planar Systems**

box\_x and box\_y define the user specified lateral dimensions of the planar membrane system. LiMBS1.0 records these dimensions as part of the membrane specification but does not calculate or optimize them. For ratio mode planar systems, the specified lateral dimensions are passed to INSANE, which determines the final integer lipid populations during membrane construction.

## 5.2 Normal Dimension (box\_z) Planar Systems

box\_z must accommodate the complete bilayer together with sufficient solvent on both sides to avoid undesired interactions with periodic images. Martini CG bilayers commonly have thickness on the order of several nanometers, with values around 4-5 nm observed for representative phospholipid membranes. The required solvent region depends on the membrane composition, interaction cutoffs, and any additional solutes. For simple CG bilayers, a total box\_z on the order of 8-10 nm may provide a useful initial setup, but the appropriate value should be verified for the specific system.

## 5.3 Box Dimensions — Vesicle Systems

For type: vesicle, approximately equal x, y, and z dimensions are generally appropriate for a spherical vesicle. The simulation box should be large enough to contain the complete outer vesicle surface together with sufficient solvent separation from its periodic images.

```
box_dimension ≥ vesicle_outer_diameter + 2 x solvent_buffer
```

The .tsi geometry file used by TS2CG PLM defines the triangulated membrane surface and contains its associated box information. Therefore, the LiMBS box dimensions, .tsi geometry, and any PLM rescaling applied to the surface should be checked for consistency before membrane construction.

## 5.4 Sanity-Checking Before a Full Run

- Always dry-run first (--dry-run) and inspect the generated command before executing INSANE or TS2CG — this is the cheapest place to catch a box/composition mismatch.
- For planar systems in ratio mode, INSANE, not LiMBS, determines the final integer lipid populations during membrane packing based on the supplied composition and box dimensions. If the resulting membrane appears under- or over-packed, inspect and adjust the lateral box dimensions rather than changing ratios solely to control lipid number.
- For vesicle systems, leaflet ratios are passed to TS2CG through the [Lipids List] section of input.str, TS2CG determines the resulting lipid populations during construction of curved membrane surface.

## 6. Tail Fragment Naming

### 6.1 Default Convention

```
#T<carbons>_<double_bonds>
```

Examples:

- #T16 — 16-carbon saturated
- #T18\_1 — 18-carbon, one double bond
- #T18\_2 — 18-carbon, two double bonds
- #T20\_4 — 20-carbon, four double bonds

### 6.2 With Double-Bond Position

```
#T<carbons>_<double_bonds>_D<position>
```

Examples:

- #T18\_1\_D9 — Oleoyl — delta-9 (POPC)
- #T18\_1\_D11 — Vaccenoyl — delta-11 (PVPE)
- #T18\_2\_D9\_D12 — Linoleoyl

## 7. Solvent and Salt

### 7.1 Solvent Species

| Field     | Description      | Example                                                  |
|-----------|------------------|----------------------------------------------------------|
| sol:TIP3P | Atomistic water  | TIP3P=AA [ O ] ]                                         |
| sol:W     | MARTINI CG water | W=CG [ [OOOO:^W]  <br>[ ^W ] ]                           |
| sol:WW    | SCG water        | WW=SCG [<br>[OOOOOOOO:^WW]  <br>[ ^WW ]   [ ^SOLVENT ] ] |

### 7.2 Salt Handling Rules

The salt: field is optional. If omitted, the LiMBS specification does not explicitly define added salt.

- Concentration > 0 → salt species block required
- Concentration = 0M → no salt species block
- Generate only the salt named in the header

CaCl2 example (CG):

```
salt:0.15M CaCl2
CaCl2=CG| [
[Ca2+:^Ca] [Cl-:^Cl] [Cl-:^Cl]
|
[ ^Ca ] [ ^Cl ] [ ^Cl ]
]
```

## 8. Validation Rules

### 8.1 Required Checks

The LiMBS parser performs syntactic and semantic checks before a specification is accepted. Important checks include:

- The specification must begin with the exact version token LiMBS1.0.
- Exactly one |++| separator must separate the system block from the species-definition block.
- Required system fields, including resolution, lipid metadata, leaflet composition, membrane type, box dimensions, and solvent, must be present.
- The resolution must be one of AA, CG, and SCG.
- The membrane type must be planar or vesicle.
- The leaflet mode must be count or ratio. In count mode, leaflet values must be positive integers; in ratio mode, leaflet values must be positive.



- Every lipid appearing in a leaflet must be present in the lipid metadata block.
- Every leaflet species must have a corresponding species block definition.
- Species definitions must be unique, and species names must match consistently across metadata, leaflet, and species-definition blocks.
- Every referenced fragment must be defined.
- Species-definitions brackets must be balanced.
- Box dimensions must contain three positive values and use unit of nm.
- Solvent compatibility is checked according to molecular resolution. For example, CG systems require W, whereas SCG systems require WW.
- Unsupported species-block resolutions are reported as errors.
- The parser reports warnings when a lipid species block uses a resolution different from the system resolution.
- The parser reports warnings when a declared solvent or required salt/ion representation cannot be located in the species definitions.

## 8.2 Common Errors and Fixes

| Code        | Meaning                                         | Common cause                                                                      |
|-------------|-------------------------------------------------|-----------------------------------------------------------------------------------|
| LMB-E000    | Empty input or empty block                      | File is blank or  ++  missing                                                     |
| LMB-E001    | Does not begin with LiMBS1.0                    | First line wrong or file truncated                                                |
| LMB-E002    | Not exactly one  ++  separator                  | Duplicate  ++ , or missing entirely                                               |
| LMB-E003    | Duplicate system field                          | Two {lipid} blocks, or field appears twice                                        |
| LMB-E004    | Invalid resolution                              | rsln: CG(space after colon) instead of rsln:CG                                    |
| LMB-E010/11 | Invalid lipid metadata                          | Missing commas, wrong format, unrecognized text                                   |
| LMB-E012    | Duplicate lipid in metadata                     | Same name listed twice inside{}                                                   |
| LMB-E020    | Invalid leaflet block                           | Wrong syntax                                                                      |
| LMB-E024    | Duplicate lipid in the same leaflet composition | Same lipid identifier is listed more than once within the same -u{} or -l{} block |
| LMB-E040    | Invalid box block                               | Space before bracket, wrong unit, missing values                                  |
| LMB-E060/61 | Invalid salt field                              | Wrong format or missing species name                                              |
| LMB-E071    | Invalid species definition                      | Missing = or resolution keyword                                                   |
| LMB-E072    | Duplicate species                               | Same name defined twice after  ++                                                 |

|             |                                          |                                                                      |
|-------------|------------------------------------------|----------------------------------------------------------------------|
| LMB-E080/81 | Unbalanced brackets                      | Missing [or] in species body                                         |
| LMB-E107    | Box unit not nm                          | Angstrom or other unit used                                          |
| LMB-E108    | Invalid box dimension                    | One or more box dimensions are zero or negative                      |
| LMB-E116    | Leaflet lipid missing from metadata      | Lipid in leaflet:count not in {} block                               |
| LMB-E117    | Invalid leaflet ratio sum                | Leaflet:ratio values in the upper or lower leaflet do not sum to 1.0 |
| LMB-E120    | Leaflet lipid missing species definition | Lipid in leaflets has no matching species block                      |
| LMB-E140    | Undefined fragment used                  | #TailA used but {#TailA=..} not defined                              |
| LMB-E160    | Undefined semantic group                 | ^^HEAD used but {^^HEAD=..} not defined                              |
| LMB-W121    | Solvent has no species definition        | sol:W declared but W=CG[...] missing                                 |
| LMB-W122    | Salt ion not found in species blocks     | NaCl declared but no Na/Cl bead labels found                         |
| LMB-W130    | Species resolution mismatch              | DPPC defined as AA but system resolution is CG                       |
| LMB-W151    | CG species have only one layer           | Missing   between canonical and mapping layers                       |
| LMB-W153    | SCG species have fewer than three layers | Missing   between layers                                             |
| LMB-E180    | Invalid solvent for CG resolution        | CG system specifies a solvent other than W                           |
| LMB-E181    | Invalid solvent for SCG resolution       | SCG system specifies a solvent other than WW                         |
| LMB-W180    | Unrecognized AA solvent                  | AA system specifies a solvent other than TIP3P, TIP4P, SPC, or SPCE  |

### 8.3 Canonical Ordering

- Lipid names sorted alphabetically in metadata and leaflet
- Upper leaflet before lower leaflet
- Species blocks sorted alphabetically after |++|
- All units explicit (nm for box, M for salt)

## 9. Representative LiMBS Examples

### 9.1 DPPC Planar Bilayer — Atomistic

```

LiMBS1.0 ||
rsln:AA ||
lsc:1 ||
uls:1 ||
lls:1 ||
{DPPC:tails=16:0-16:0,head=PC,charge=0.0} ||
leaflet:count{-u{DPPC:158}-l{DPPC:158}} ||
type:planar ||
box:[10.0,10.0,8.0]nm ||
sol:TIP3P ||
salt:0.15M NaCl
|++|
DPPC=AA| [
{#T16=CCCCCCCCCCCCCCCC}
[N] (C) (C) (C) CCOP(=O) (O) OCCO(=O) [#T16] CO(=O) [#T16]
]
|+|
NaCl=AA| [
[Na+] [Cl-]
]
|+|
TIP3P=AA| [
[O]
]

```

## 9.2 System-Block DOPC/POPA/POPE Planar Bilayer — CG (Count Mode)

```

LiMBS1.0 ||
rsln:CG ||
lsc:3 ||
uls:3 ||
lls:3 ||
{DOPC:tails=18:1-18:1,head=PC,charge=0.0;
POPA:tails=16:0-18:1,head=PA,charge=-1.0;
POPE:tails=16:0-18:1,head=PE,charge=0.0} ||
leaflet:count{-u{DOPC:40,POPA:10,POPE:10}-l{DOPC:40,POPA:10,POPE:10}} ||
type:planar ||
box:[12.0,12.0,9.0]nm ||
sol:W ||
salt:0.15M NaCl

```

## 9.3 System-Block DOPC/POPA/POPE Planar Bilayer — CG (Ratio Mode)

The same composition as 9.2, expressed with `leaflet:ratio` instead of `leaflet:count`. INSANE will determine the actual lipid counts from `box:[12.0,12.0,9.0]nm` during packing:

```

LiMBS1.0 ||
rsln:CG ||
lsc:3 ||
uls:3 ||
lls:3 ||
{DOPC:tails=18:1-18:1,head=PC,charge=0.0;
POPA:tails=16:0-18:1,head=PA,charge=-1.0;
POPE:tails=16:0-18:1,head=PE,charge=0.0} ||
leaflet:ratio{-u{DOPC:0.67,POPA:0.165,POPE:0.165}-

```

```
l{DOPC:0.67,POPA:0.165,POPE:0.165}} ||
type:planar ||
box:[12.0,12.0,9.0]nm ||
sol:W ||
salt:0.15M NaCl
```

## 9.4 System-Block POPC/CHOL Vesicle — CG

```
LiMBS1.0 ||
rsln:CG ||
lsc:2 ||
uls:2 ||
lls:2 ||
{CHOL:tails=sterol,head=CHOL,charge=0.0;
POPC:tails=16:0-18:1,head=PC,charge=0.0} ||
leaflet:ratio{-u{CHOL:0.30,POPC:0.70}-l{CHOL:0.30,POPC:0.70}} ||
type:vesicle ||
box:[20.0,20.0,20.0]nm ||
sol:W ||
salt:0.15M NaCl
```

## 9.5 System-Block E. coli Outer Membrane — CG (Asymmetric, Count Mode)

```
LiMBS1.0 ||
rsln:CG ||
lsc:4 ||
uls:1 ||
lls:3 ||
{LPS:tails=mixed,head=K12_core_LPS,charge=-8.0; POPE:tails=16:0-
18:1,head=PE,charge=0.0;POPG:tails=16:0-18:1,head=PG,charge=-
1.0;PVCL2:tails=16:0-18:1-16:0-18:1,head=CL,charge=-2.0}
||
leaflet:count{
-u{LPS:294}
-l{POPE:768,POPG:43,PVCL2:43}
} ||
type:planar ||
box:[23.0,23.0,15.0]nm ||
sol:W ||
salt:0.15M CaCl2
```

## 10. Quick Reference

### Delimiters

| Field | Description                               | Example                         |
|-------|-------------------------------------------|---------------------------------|
|       | Separates system header fields            | rsln:CG                         |
| ++    | Separates system block from species block | system  ++  species             |
| +     | Separates individual species blocks       | DPPC=CG [...]  + <br>W=CG [...] |
|       | Separates layers within a species block   | layer1   layer2   layer3        |

## Geometry vs Leaflet Mode

| Field        | Description                                                                                             | Valid leaflet modes |
|--------------|---------------------------------------------------------------------------------------------------------|---------------------|
| type:planar  | Flat bilayer; CG planar systems can be translated to INSANE input using the LiMBS_INSANE interface.     | count or ratio      |
| type:vesicle | Spherical vesicle; CG vesicle systems can be translated to TS2CG input using the LiMBS_TS2CG interface. | ratio only          |

## Resolution Layers

| Field | Description | Example                                    |
|-------|-------------|--------------------------------------------|
| AA    | 1 layer     | SMILES structure                           |
| CG    | 2 layers    | chemical mapping   bead sequence           |
| SCG   | 3 layers    | chemical   bead sequence   semantic groups |

## Most Common Mistakes

- Space after field key: rsln: CG instead of rsln:CG
- Missing || on a header field
- Using leaflet:count for a vesicle (vesicles require leaflet:ratio)
- Assuming leaflet:ratio on a planar system fixes the final lipid count — it does not; INSANE determines it from the box (Section 4.3)
- Lipid name mismatch between metadata and species block
- Salt block present when concentration is 0M
- Forgetting |+| between species blocks

## 11. Using the Parser Scripts

The LiMBS repository includes three Python scripts. The LiMBS1.0 reference implementation requires Python 3 and was tested with Python 3.11.15. Place your LiMBS notation in a plain text file (e.g. system.txt) before running any script.

| Field             | Description                                                                | Example                                   |
|-------------------|----------------------------------------------------------------------------|-------------------------------------------|
| LiMBSv1_parser.py | Core parser — validation, summary, JSON export, canonicalization           | python<br>LiMBSv1_parser.py<br>system.txt |
| LiMBS_INSANE.py   | LiMBS_INSANE.py — converts CG planar LiMBS to INSANE input and runs INSANE | python<br>LiMBS_INSANE.py<br>system.txt   |

| Field          | Description                                                                  | Example                                |
|----------------|------------------------------------------------------------------------------|----------------------------------------|
| LiMBS_TS2CG.py | LiMBS_TS2CG.py — converts CG vesicle LiMBS to TS2CG input.str and runs TS2CG | python<br>LiMBS_TS2CG.py<br>system.txt |

### 11.1 LiMBSv1\_parser.py — Core Parser

This is the foundation script. It reads a LiMBS text file, validates it against all syntax rules, and can output a human-readable summary, a canonical form, or a JSON export.

#### Command-line flags

| Field           | Description                                             | Example                                                   |
|-----------------|---------------------------------------------------------|-----------------------------------------------------------|
| (none)          | Validate and print a parsed summary to the terminal     | python<br>LiMBSv1_parser.py<br>system.txt                 |
| --validate-only | Validate only — exit 0 on pass, exit 1 on failure       | python<br>LiMBSv1_parser.py<br>system.txt --validate-only |
| --canonical     | Print the canonical LiMBS string to standard output     | python<br>LiMBSv1_parser.py<br>system.txt --canonical     |
| --json          | Print the full parsed system as JSON to standard output | python<br>LiMBSv1_parser.py<br>system.txt --json          |
| --quiet         | Suppress the summary — only print pass/fail             | python<br>LiMBSv1_parser.py<br>system.txt --quiet         |

#### Step 1 — Validate and Review the Summary

Run without any flags to validate and see a full parsed summary:

```
python LiMBSv1_parser.py system.txt
```

On a valid file the terminal output will look like this:

```
LiMBS1.0 parsed system
Version : 1.0
Resolution : CG
Lipid sp. count : 1
U leaflet species: 1
L leaflet species: 1
Type : planar
Box : [10.0, 10.0, 8.0] nm
Solvent : W
Salt : 0.15 M NaCl
Leaflet mode : count
Upper leaflet : {'DPPC': 56}
Lower leaflet : {'DPPC': 56}
```

```
Lipid metadata : ['DPPC']
Species defs : ['DPPC', 'NaCl', 'W']
DPPC : CG, 2 layer(s)
Fragments defined: ['TailA', 'TailB']
Fragments used : ['TailA', 'TailB']
Validation passed.
```

Use `--validate-only` for scripting or CI pipelines — it prints only pass/fail and sets the exit code:

```
python LiMBSv1_parser.py system.txt --validate-only
# exit code 0 = passed, exit code 1 = failed
```

On failure, an error code and message are printed:

```
LMB-E002: LiMBS string must contain exactly one |++| separator
# or
LMB-E120: Species 'DPPC' in upper leaflet has no species block defined.
```

### Step 2 — Canonicalize (Optional)

The `--canonical` flag outputs a normalized, alphabetically sorted version of a valid LiMBS string. This is useful for comparing two systems that describe the same membrane, or for storing a standardized representation.

```
python LiMBSv1_parser.py system.txt --canonical
```

To save the canonical form to a file:

```
python LiMBSv1_parser.py system.txt --canonical > system_canonical.txt
```

**NOTE:** Canonicalization only succeeds in a valid file. If the file fails validation, the script exits with an error before producing canonical output.

### Step 3 — Export JSON (Optional but Recommended)

The `--json` flag outputs the full parsed system as a structured JSON object. This is the format consumed internally by `LiMBS_INSANE.py` and `LiMBS_TS2CG.py`.

```
python LiMBSv1_parser.py system.txt --json
```

To save the JSON to a file:

```
python LiMBSv1_parser.py system.txt --json > system.json
```

The JSON output includes the resolution, membrane type, box, solvent, salt, leaflet composition (counts or ratios, exactly as written), lipid metadata, species block details, and any warnings or errors.

## 11.2 LiMBS\_INSANE.py — INSANE Interoperability Interface

Converts a validated CG planar LiMBS file into an INSANE command and optionally executes it to produce membrane.gro and membrane.top. Requires LiMBSv1\_parser.py and access to a compatible insane.py may be supplied using `--insane-script`.

**CRITICAL:** Only works with CG resolution and type: planar. Using AA, SCG, or type: vesicle will raise a compatibility error before any file is written. Both leaflet: count and leaflet: ratio is accepted for type: planar.

### Command-line flags

| Field             | Description                                        | Example                   |
|-------------------|----------------------------------------------------|---------------------------|
| input             | LiMBS .txt file (required positional argument)     | system.txt                |
| --dry-run         | Print the INSANE command without executing it      | --dry-run                 |
| --write-json PATH | Save the intermediate parsed JSON to a file        | --write-json out.json     |
| --from-json       | Start from a previously saved LiMBS JSON file      | --from-json out.json      |
| --insane-script   | Path to insane.py (default: insane.py)             | --insane-script insane.py |
| --python-exe      | Python executable to run INSANE (default: python3) | --python-exe python       |
| --out-gro         | Output .gro filename (default: membrane.gro)       | --out-gro bilayer.gro     |
| --out-top         | Output .top filename (default: membrane.top)       | --out-top bilayer.top     |
| --rand            | INSANE randomisation seed (default: 0.1)           | --rand 0.5                |
| --solr            | INSANE solute rotational freedom (default: 0.5)    | --solr 0.3                |

### Step 1 — Validate the File First

```
python LiMBSv1_parser.py system_CG.txt --validate-only
```

### Step 2 — Dry-Run to Inspect the Generated Command

```
python LiMBS_INSANE.py system_CG.txt --dry-run
```

This works for both leaflet:count and leaflet:ratio systems — the printed `-u/-l` values will show absolute counts or relative proportions respectively, exactly as they appear in the LiMBS file.

### Step 3 — Run INSANE to Produce Membrane Files

```
python LiMBS_INSANE.py system_CG.txt --insane-script insane.py
```

Expected terminal output on success (count mode shown):

```
Parsing LiMBS file: system_CG.txt
Validation: PASSED
```



INSANE compatibility: OK

Generated INSANE command:

```
python3 insane.py -u DPPC:56 -l DPPC:56 -x 10.0 -y 10.0 -z 8.0 -sol W -salt 0.15 ...
```

Running INSANE...

INSANE completed successfully.

Coordinates : membrane.gro

Topology : membrane.top

In ratio mode, the -u/-l values printed in the generated command are fractional (e.g. -u POPC:0.70 -u CHOL:0.30), and INSANE resolves them into actual lipid counts during its own packing step — the counts reported in the resulting membrane.top reflect what INSANE actually placed, not a number computed by LiMBS.

| OUTPUT       | Description                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| membrane.gro | GROMACS coordinate file of the INSANE-built membrane system; subsequent topology/force-field verification and MD preparation are required |
| membrane.top | GROMACS topology file listing lipid and solvent molecule counts                                                                           |

### 11.3 LiMBS\_TS2CG.py — TS2CG Interoperability Interface

LiMBS\_TS2CG.py converts a validated CG vesicle LiMBS specification into a TS2CG input.str file and can optionally execute the TS2CG PLM and PCG stages to construct the vesicle.

The workflow requires LiMBSv1\_parser.py, a TS2CG triangulated-surface geometry file such as sphere.tsi, a compatible Martini lipid library such as Martini3.LIB, and an explicitly supplied area-per-lipid (APL) value for each lipid included in vesicle.

**CRITICAL:** Only works with CG resolution and type: vesicle. The TS2CG executable must be available in the user's PATH for full execution.

#### Command-line flags

| Field               | Description                                                                                                              | Example                 |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------|
| input               | LiMBS .txt file (required positional argument)                                                                           | CG_vesicle.txt          |
| --validate-only     | Parse and validate the LiMBS specification and check TS2CG compatibility — write nothing                                 | --validate-only         |
| --write-str-only    | Generate input.str only; do not run TS2CG                                                                                | --write-str-only        |
| --apl LIPID=VALUE   | Area per lipid in nm <sup>2</sup> for a lipid used by TS2CG. Supply an APL value for every lipid present in the vesicle. | --apl DPPC=0.64         |
| --dry-run           | Generate input.str and print the TS2CG commands without executing them                                                   | --dry-run               |
| --tsi FILE          | TS2CG triangulated-surface geometry file (default: sphere.tsi)                                                           | --tsi sphere.tsi        |
| --lib FILE          | Compatible Martini lipid library                                                                                         | --lib Martini3.LIB      |
| --str-file FILE     | Output TS2CG composition filename (default: input.str)                                                                   | --str-file myinput.str  |
| --out-name NAME     | Output prefix (default: vesicle)                                                                                         | --out-name myvesicle    |
| --bilayer-thickness | PLM bilayer thickness in nm (default: 3.0)                                                                               | --bilayer-thickness 3.8 |
| --rescale X Y Z     | PLM surface rescaling factors (default: 4.8 4.8 4.8)                                                                     | --rescale 5.0 5.0 5.0   |
| --mashno INT        | PLM Mashno value (LiMBS wrapper default: 4)                                                                              | --mashno 6              |
| --bond-length       | PCG bond length in nm (LiMBS wrapper default: 0.2)                                                                       | --bond-length 0.25      |

The APL value is a TS2CG builder-specific input and is not assigned automatically by LiMBS1.0. Appropriate values should be selected for the lipid model and membrane conditions being used.

#### Step 1 — Validate the Vesicle File

Before generating any TS2CG files, validate the LiMBS specification:

```
python LiMBSv1_parser.py vesicle_CG.txt --validate-only
```

For a compatible system, the output should include:

```
Validation: PASSED
TS2CG compatibility: OK
```

### ***Step 2 — Write input.str Only (Inspect Before Running TS2CG)***

Generate the TS2CG composition file without running TS2CG:

```
python LiMBS_TS2CG.py vesicle_CG.txt --apl DPPC=0.64 --write-str-only
```

Inspect the generated file:

```
vi input.str
```

For the DPPC vesicle, the generated input.str is:

```
[Lipids List]
;LipidName  RatioUp  RatioDown  Area/Lipid
Domain 0
DPPC  1 1 0.64
End
```

### ***Step 3 — Dry-Run to Inspect TS2CG Commands***

Before executing TS2CG, a dry run is recommended:

```
python LiMBS_TS2CG.py vesicle_CG.txt --apl DPPC=0.64 --dry-run
```

The script generates input.str and prints the PLM and PCG commands but does not execute them.

A typical command preview is:

```
TS2CG PLM -TSfile sphere.tsi -bilayerThickness 3.0 -rescalefactor 4.8 4.8 4.8 -Mashno 4
TS2CG PCG -dts point -str input.str -Bond length 0.2 -LLIB Martini3.LIB -defout vesicle
```

The PLM geometry parameters are downstream TS2CG construction settings. They are configurable through the command-line options and are not encoded as core LiMBS membrane descriptors.

### ***Step 4 — Run the Complete TS2CG Workflow***

After inspecting the generated files and commands, execute the complete workflow:

```
python LiMBS_TS2CG.py vesicle_CG.txt --apl DPPC=0.64
```

For the tested DPPC vesicle, representative output is:

```
Parsing LiMBS file: vesicle_CG.txt
Validation: PASSED
Parsed LiMBS vesicle system:
Resolution : CG
Geometry : vesicle
Box : [30.0, 30.0, 30.0] nm
Solvent : W
Salt : 0.15 M NaCl
Leaflet mode: ratio
Outer (upper) : {'DPPC': 1.0}
Inner (lower) : {'DPPC': 1.0}

TS2CG compatibility: OK
DPPC: 0.64 nm2
[TS2CG input.str]
Wrote: input.str

[Step 1] TS2CG PLM
TS2CG PLM -TSfile sphere.tsi -bilayerThickness 3.0 -rescalefactor 4.8 4.8
4.8 -Mashno 4

[Step 2] TS2CG PCG
TS2CG PCG -dts point -str input.str -Bondlength 0.2 -LLIB Martini3.LIB -
defout vesicle

TS2CG completed successfully.
Output prefix : vesicle
```

| OUTPUT      | Description                                      |
|-------------|--------------------------------------------------|
| vesicle.gro | GROMACS coordinate file of the assembled vesicle |
| vesicle.top | GROMACS topology file                            |
| input.str   | TS2CG composition file (kept for reference)      |

## 11.4 Recommended Workflow — Step by Step

### Step 1 — Write the LiMBS Notation

Draft the system in a plain text file (e.g. system.txt), following the field reference in Section 2 and the leaflet-mode guidance in Section 4.

### Step 2 — Validate

Run LiMBSv1\_parser.py with no flags and read the summary carefully. Fix any reported errors before moving on.

```
python LiMBSv1_parser.py system.txt
```

### **Step 3 — Archive the Canonical Form (Optional)**

Save `--canonical` output as the version-controlled, archived representation of the system.

```
python LiMBSv1_parser.py system.txt --canonical > system_canonical.txt
```

### **Step 4 — Export JSON (Optional)**

Save `--json` output for inspection or for passing directly to the exporter scripts with `--from-json`.

```
python LiMBSv1_parser.py system.txt --json > system.json
```

### **Step 5 — Dry-Run the Exporter**

Before executing anything, dry-run `LiMBS_INSANE.py` or `LiMBS_TS2CG.py` as appropriate and verify the generated command — this is the point at which a box/mode mismatch (Sections 4–5) is easiest to catch.

```
python LiMBS_INSANE.py system.txt --dry-run  
# or  
python LiMBS_TS2CG.py system.txt --dry-run
```

### **Step 6 — Run the Exporter**

Execute the exporter to produce the final `.gro` and `.top` (and, for vesicles, `input.str`).

**NOTE:** Always validate with `LiMBSv1_parser.py` before running `LiMBS_INSANE.py` or `LiMBS_TS2CG.py`. The exporter scripts also validate internally, but running the parser first gives a clearer error message if something is wrong.

## 12. What LiMBS Generates vs. What INSANE/TS2CG Generates

It is easy to lose track of which files come from the LiMBS toolchain itself, and which come from the external packing engines it drives. This section separates the two.

### 12.1 Files Produced by LiMBS Scripts

| File                 | Produced by                                                 | Notes                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| system_canonical.txt | LiMBSv1_parser.py --canonical                               | Normalized, alphabetically sorted LiMBS notation. Intended mainly for archival, comparison, and reproducibility purposes.                                                                                                                             |
| system.json          | LiMBSv1_parser.py --json (or -write-json via the exporters) | Full parsed representation of the system including resolution, box information, solvent, salt settings, leaflet definitions, lipid counts or ratios as supplied by the user, and species definitions.                                                 |
| input.str            | LiMBS_TS2CG.py                                              | TS2CG composition file generated by the LiMBS-TS2CG interface. It contains the upper and lower leaflet composition together with the TS2CG area-per-lipid values supplied by the user. The completed input.str is subsequently used by the TS2CG PCG. |

### 12.2 Files Produced by the External Packing Engines

| File                                                   | Produced by      | Notes                                                                                                                                                                                     |
|--------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| membrane.gro (or custom name specified with --out-gro) | INSANE           | Coordinates for the packed planar membrane. LiMBS prepares and supplies the command-line arguments while INSANE performs the packing and writes this file.                                |
| membrane.top (or custom name specified with --out-top) | INSANE           | Topology containing the final molecule counts generated during system generation. In ratio mode, this is the where the actual integer number of lipids placed by INSANE become available. |
| vesicle.gro (or custom name specified --out-name)      | TS2CG (PCG step) | Coordinates for the packed vesicle or curved membrane generated by the PCG stage.                                                                                                         |
| vesicle.top                                            | TS2CG (PCG step) | Topology generated for the membrane system by the PCG stage.                                                                                                                              |

**NOTE:** In ratio mode especially, the .top file is the authoritative record of the integer lipid counts present in the constructed system. LiMBS preserves the user-specified leaflet ratios but does not convert those ratios into final integer molecule counts. INSANE or TS2CG determines the actual numbers during membrane construction.

In count mode, the requested lipid numbers are already explicitly defined by the user, although the final topology should still be checked to confirm that the constructed system matches the intended composition.

## 13. What Is Still Required Before MD?

After a LiMBS specification is translated through the corresponding interface, the external builder generates the coordinate and topology file, but the resulting system should not automatically be considered ready for production molecular dynamics. The topology, composition, charge, structure, and equilibration behavior should be checked carefully.

### 13.1 Topology and Force Field

- Confirm that all #include statements in the .top file resolve to the correct force-field and molecular .itp files required for the selected model.
- Confirm that the force-field version used for the simulation is compatible with the lipid definitions used during system construction.
- Check that the lipid, solvent, and ion molecule counts in the [ molecules ] section of topology are consistent with the intended system.
- For ratio-mode systems, verify the final integer lipid counts because the packing engine determines these numbers during construction.

### 13.2 System Neutrality and Ions

- If INSANE is used with salt or ion options, verify the final ion composition rather than assuming that the requested concentration automatically produced the intended final system.
- Confirm the total system charge after membrane construction. In GROMACS workflow, preprocessing the system with gmx grompp provides an additional check because GROMACS reports a warning when the complete topology has a non-zero total charge.
- The TS2CG PCG stage constructs the membrane coordinates and topology but does not itself perform the complete solvation and salt-adjustment procedure.

### 13.3 Structural Sanity Checks

- Visually inspect the .gro file using a molecular visualization program such as VMD or PyMOL.
- Check for obvious structural problems, including overlapping lipids, severe steric clashes, lipids oriented incorrectly, solvent molecules inside the hydrophobic membrane core, incomplete membrane coverage, unexpected holes and so on.

### 13.4 Energy Minimization

- Perform an energy minimization step before beginning molecular dynamics.
- A steepest-descent minimization is commonly used to remove severe steric clashes introduced during initial membrane packing.
- Check the minimization log for a converged, sufficiently low maximum force before proceeding.

### **13.5 Equilibration**

- The constructed system should undergo an appropriate equilibration protocol consistent with the selected force field and membrane model before production simulation.
- LiMBS1.0 does not prescribe a specific equilibration protocol. Parameters such as ensemble selection, thermostat, barostat, pressure, temperature, restraints, timestep, and equilibration duration must be selected according to the force field and simulation protocol and are not specified by LiMBS1.0.

### **13.6 Anything LiMBS Does Not Add**

- Embedded or peripheral proteins, peptides, or other non-lipid solutes are not part of the LiMBS notation and must be inserted and re-solvated separately.
- Restraints, pulling code, or enhanced-sampling setup (umbrella sampling, metadynamics, etc.) are configured entirely outside the LiMBS/INSANE/TS2CG toolchain.
- Production-run MD parameters (integrator, thermostat/barostat choice, cutoffs, run length) are a separate step and are not generated by any of the three scripts in Section 11.

### **13.7 Production**

- Once minimization and equilibration are validated, the system can proceed to a production molecular dynamics simulation.